

TED UNIVERSITY

CMPE 491 Senior Project

BioAuth Project Proposal

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1. Overview

BioAuth is a physiological liveness-aware biometric authentication system designed to improve the robustness of remote identity verification processes. The system integrates facial similarity verification with remote photoplethysmography (rPPG) based pulse signal analysis to determine whether the presented identity corresponds to a biologically live person.

2. Problem Statement

Modern digital onboarding solutions commonly rely on visual behavior-based liveness detection methods such as blinking or head-movement challenges. However, recent developments in generative AI and deepfake technologies enable the creation of highly realistic synthetic facial animations that may bypass conventional verification pipelines. This creates a need for authentication mechanisms leveraging intrinsic physiological signals.

3. Objectives

- Develop a hybrid biometric authentication pipeline combining face recognition and physiological liveness detection.
- Extract and analyze rPPG signals from facial regions of interest in real time.
- Evaluate robustness against spoofing attacks including print, replay, and AI-generated portrait animation attacks.
- Implement a fusion-based decision mechanism integrating visual and physiological confidence metrics.

4. Methodology

1. Face detection and facial landmark extraction
2. Region-of-interest selection for pulse signal monitoring
3. rPPG signal extraction using CHROM and POS algorithms
4. Temporal filtering and frequency domain pulse estimation
5. Liveness scoring and multi-factor authentication decision fusion

5. Expected Outputs

- Hybrid physiological-visual biometric authentication prototype
- Controlled spoof attack evaluation dataset
- Real-time signal visualization and monitoring dashboard
- Mobile-compatible authentication workflow demonstration

6. Project Timeline

Phase 1 – Literature Review and Requirement Analysis

Relevant studies on physiological liveness detection, remote photoplethysmography, and deepfake-resistant biometric systems will be analyzed. Modern cybersecurity threats such as deepfake, replay, injection attacks, and device/source integrity risks will also be examined. System requirements, threat models, and evaluation criteria will be defined.

Phase 2 – Core Authentication Pipeline Development

The biometric verification framework will be implemented, including face detection, facial landmark extraction, region-of-interest selection, and physiological signal processing modules. Security components such as input validation, secure data handling, and protection against manipulated inputs will be integrated.

Phase 3 – Spoofing Scenario Design and Experimental Evaluation

Spoof attack scenarios (print, replay, AI-generated portraits) and system-level attacks (fake webcam, emulator, injection) will be designed. The robustness of the proposed authentication system will be evaluated under these conditions.

Phase 4 – Multi-Factor Decision Fusion and System Optimization

Fusion strategies combining facial similarity, motion-based validation, physiological signals, and device/source integrity will be developed and optimized. A risk-based decision mechanism will improve robustness against biometric and system-level attacks.

Phase 5 – Final System Integration and Demonstration Preparation

All components will be integrated into a unified prototype. The system will include both biometric and cybersecurity layers, and performance results and demo materials will be prepared.